Budget Automation System - Functional Testing Report

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This document records functional testing performed on the PBAS frontend and backend. It explains how each main function works, what was verified, and what issues were observed during testing.

Report Date	2026-08-29 17:45 PKT
Frontend Repo	PBAS-Frontend, branch `main`
Backend Repo	PBAS-Backend, branch `main`
Frontend Commit Tested	`4b7727f Add separate estimates management module`
Backend Commit Tested	`b1d7857 Add payroll estimates API`
Frontend URL	`http://localhost:5173`
Backend API URL	`http://localhost:3002/api/v1`

Executive Summary

Backend

PASS

57 tests, 280 assertions, 0 failures.

Frontend

PASS WITH 15s TIMEOUT

11 test files, 18 tests passed.

Build/Lint

PASS

Production build and ESLint completed.

Core backend workflows passed. Frontend tests passed when timeout was increased to 15 seconds. With the default 5 second timeout, three frontend tests timed out, which is listed as an issue because the suite is slow and can give false failure during viva/demo preparation.

Commands Executed

Command	Purpose	Result
`npm run lint`	Frontend static code validation.	PASS No lint errors. Warning: baseline browser mapping data is old.
`npm run build`	Production build validation.	PASS Build completed. Warning: JS chunk above 500 kB.
`npm test`	Frontend automated tests with default timeout.	TIMEOUT 15 passed, 3 timed out at 5000 ms.
`npm test -- --testTimeout=15000`	Frontend automated tests with practical timeout.	PASS 11 files, 18 tests passed.
`bin/rails test`	Backend full automated test suite.	PASS 57 runs, 280 assertions, 0 failures, 0 errors.
`bin/rails test test/integration/payroll_estimates_flow_test.rb`	Focused Estimates Management backend testing.	PASS 2 runs, 16 assertions.

Module Functional Testing

Module	Main Functions	How It Works	Testing Status	Observed Issue
Authentication	Login, splash login, forgot password, reset password, email verification, protected routes.	User opens splash screen, clicks Login, enters credentials, frontend stores JWT auth state, protected route checks token and redirects unauthenticated users to login.	PASS ProtectedRoute test confirms redirect without token. Backend sessions/registrations routes available.	No functional failure observed in automated testing.
Dashboard	Display live totals, charts/cards, recent system data.	Dashboard calls backend dashboard endpoint and renders returned metrics for quick management overview.	PASS Frontend Dashboard test renders live dashboard data.	No functional failure observed.
User Management	List users, create user modal, update/delete user, lock/unlock user, role-based access.	Only users with user permissions see the module. Data is fetched from `/api/v1/users`. Create/update/delete actions go through backend API and role permissions.	PASS Passed with 15s frontend timeout. Backend user tests and access matrix passed.	SLOW TEST Default 5s frontend test timed out. AntD warning: `destroyOnClose` deprecated.
Resource Request Management	Create request, verify, approve, reject, department request tracking.	Request is submitted by permitted user, verified by admin/account staff, then approved/rejected by Head of Institute according to workflow permissions.	PASS Backend request flow tests passed. Frontend request test passed with 15s timeout.	SLOW TEST Default 5s frontend test timed out. AntD warnings for Card/Table/Modal/Space deprecated props.
Budget Management	Budget heads, categories, yearly budget records, allocation search/grouping.	Budgets are grouped by year and searchable across loaded records. Backend provides budget CRUD plus structure endpoints for heads/categories.	PASS Backend budget tests passed. Frontend budget test passed with 15s timeout.	SLOW TEST Default 5s frontend test timed out. AntD Card `bordered` warning appears.
Payroll Management	Staff Master, import payroll sheet staff, payroll cycle generation, approval/rejection, payment marking, payslip view/print, annual allocation/archive.	Staff records keep BPS, basic pay, allowance, deductions, increment date/rate. Payroll cycle generates monthly run items from staff data. Approver changes status, payment action records paid amount. Payroll estimates were removed from this module.	PASS Backend payroll, payroll staff, and payroll payment tests passed.	No backend failure observed. Frontend has no dedicated Payroll component test, so browser-level payroll UI coverage should be added.
Estimates Management	Create/list/delete payroll estimates, fetch staff from Payroll Staff Master, formula preview and save.	User selects staff and year. System calculates: basic pay x 4 months, basic pay x next 8 months, increment rate x 6 months, then total estimated salary. Backend saves in `payroll_estimates`.	PASS Backend focused test verifies create/list/delete, permissions, and formula. Frontend production build confirms page compiles.	Frontend component-level test for new Estimates page is not yet present. Add one for staff selection, preview formula, save, and delete.

Module	Main Functions	How It Works	Testing Status	Observed Issue
		Same staff/year updates existing estimate instead of duplicate.		
Internal Revenue	Sheet data display, add record, edit cells, save cells, recalculate totals.	Frontend behaves like a small spreadsheet. Edited cells are sent to backend internal revenue cells endpoint and reloaded into the same positions.	PASS Three frontend tests passed. Backend route provides index, bulk_save, and import_rows.	No functional failure observed.
Reports	Period controls, charts, generated report rows, analytics view.	Reports page fetches report data and renders summary tables/charts for the selected period. Chart/report rows are generated from backend reporting data.	PASS Frontend Reports test passed. Backend reports/notifications flow test passed.	No functional failure observed.
Notifications	List notifications, mark one read, mark all read, unread badge.	Layout fetches notifications for users with notification permission. User can update read status through backend notification endpoints.	PASS Frontend notification test passed. Backend reports/notifications flow test passed.	No functional failure observed.
Audit Logs	View audit records, create audit record for key actions.	Backend records action, entity type, entity id, user, IP, user agent, and metadata. Audit page displays records to permitted roles.	PASS Frontend audit log render test passed. Backend audit route available and broader tests passed.	No functional failure observed.
Settings/Profile	Profile management, role-based settings access.	User opens settings from avatar/header. Backend profile flow validates profile update access.	PASS Backend profile integration test passed.	No frontend dedicated settings test listed.
Access Control	Permission-based sidebar, protected pages, role matrix.	Frontend uses user permission list to show/hide modules. Backend enforces permissions again with `require_permission!`, so hidden UI is not the only security layer.	PASS Frontend AppLayout/AccessPages tests and backend role access matrix passed.	No functional failure observed.

Detailed Issue Log

ID	Area	Issue	Impact	Recommended Fix
FT-01	Frontend Tests	`npm test` default 5000 ms timeout fails for Users, Requests, and BudgetManagement tests.	Can confuse examiner/developer because tests look failed even though they pass with realistic timeout.	Set Vitest global `testTimeout: 15000` or optimize those tests/components.
FT-02	Frontend UI Library	Ant Design deprecation warnings: `Card bordered`, `Modal destroyOnClose`, `Table pagination.position`, `Space direction`.	Not breaking now, but future AntD upgrade may remove these props.	Replace with current AntD props: `variant`, `destroyOnHidden`, `pagination.placement`, and `orientation` where applicable.
FT-03	Frontend Build	Vite warns that JS chunk is larger than 500 kB after minification.	App still builds, but initial load can be slower on low-end systems.	Add route-level lazy loading/code splitting for large modules such as Reports, Payroll, Internal Revenue.
FT-04	Frontend Dependencies	`baseline-browser-mapping` data is more than two months old.	No immediate functional break, but browser compatibility guidance may be stale.	Run `npm i baseline-browser-mapping@latest -D` during dependency maintenance.
FT-05	Estimates Management	Backend flow is tested, but no frontend unit test exists for the new Estimates page.	Future UI change could break staff selection or preview without automated detection.	Add Vitest test that mocks `listPayrollEstimates`, selects staff, verifies formula preview, saves, and deletes.
FT-06	Payroll UI	Payroll backend is tested, but frontend Payroll page has no dedicated component test in current list.	Payroll UI regressions may only be caught manually.	Add frontend tests for Staff Master import, cycle generation button flow, payment drawer, and payslip view.

Functional Formulas Verified

Function	Formula/Rule	Example From Test	Status
Payroll Estimate First 4 Months	`monthly_basic_pay * 4`	40,000 x 4 = 160,000	PASS
Payroll Estimate Next 8 Months	`monthly_basic_pay * 8`	40,000 x 8 = 320,000	PASS
Increment Addition	`increment_rate * 6`	10 x 6 = 60 in fixture test data	PASS
Total Estimated Salary	`first_four_months + next_eight_months + increment_six_months`	160,000 + 320,000 + 60 = 480,060	PASS
Annual Payroll Remaining	`allocation - paid/utilized payroll`	Payroll module no longer subtracts estimates; estimates are separate module.	PASS

Conclusion

The backend workflows are stable according to the current automated suite. The frontend builds and passes lint. Frontend automated tests need a higher timeout or optimization; with 15 seconds all existing frontend tests pass. The main functional risks are missing frontend tests for the newly added Estimates Management page and the Payroll UI. These are recommended before final viva so the examiner can see stronger testing coverage.